

Ahmet Ruçhan AVCI

Data Scientist & AI/ML Engineer

aruchanavci01@gmail.com +905359337694 İstanbul / Türkiye
ahmetruchanavci.com ahmet-ruchana Ahmet-Ruchan

PROFILE

- Research-driven and solution-oriented **Data Scientist & AI/ML Engineer** with a strong sense of ownership and a calm, professional communication style. I turn complex, ambiguous problems into measurable outcomes through statistical thinking, hypothesis-driven experimentation, and end-to-end model development. Currently pursuing an **M.S. in Data Science and Artificial Intelligence at Boğaziçi University**, bridging hands-on production experience with rigorous academic research.

EDUCATION

Boğaziçi University, *M.S. in Data Science and Artificial Intelligence* 07/2026 | İstanbul, Türkiye
A thesis-based M.S. program grounded in Data Science, Machine Learning, and research in AI.

Beykent University, *Faculty of Engineering Computer Engineering* 11/2020 – 06/2024 | İstanbul, Türkiye
Cumulative GPA: 3.64 / 4.0
- Graduated at the pinnacle of class achievement within a competitive computer engineering curriculum, recognized with **valedictorian honors** while maintaining an average GPA above 3.64 across all semesters.

Adile Mermerci High School, *Mathematics Science* 10/2015 – 07/2019 | İstanbul, Türkiye

PROFESSIONAL EXPERIENCE

Hisar Hospital Intercontinental | Full-Time 07/2025 – Present | Üsküdar, İstanbul, Turkey
Jr. Artificial Intelligence Engineer
- Owned the design and delivery of production AI features across **ML/DL/NLP + GenAI/LLM**, from prototyping and evaluation to API release and enterprise integration into ERP/HBYS clinical workflows.
- Shipped Python/**FastAPI** services for clinical automation: an **OCR pipeline processing ≥300 documents/day** and a **Speech-to-Text workflow** used by ≥5 doctors/day (≥3 minutes each).
- Designed and implemented an end-to-end **RAG system** (ingestion → chunking → embeddings → retrieval → grounded generation) using **LangChain/LangGraph**, Vector DB, and LLM APIs over a 150-document curated knowledge base; iterated on retrieval quality via **error analysis and prompt/chunking/top-k tuning** to reduce hallucinations in production.
- Implemented backend data layer with **PostgreSQL / MS SQL Server / MongoDB** and added **Redis caching** for low-latency access.
- Owned containerized deployment and lifecycle (**Docker, Kubernetes, GitLab CI/CD**) and supported flexible LLM serving paths (**vLLM, llama.cpp, MLX**) with **MLOps-ready** patterns (versioning, monitoring, reproducibility) for stable production operation.

TÜBİTAK BİLGEM, *Intern Machine Learning Engineer* 06/2023 – 09/2023 | Gebze, Kocaeli, Turkey
- Worked at **B3LAB (Cloud Computing & Big Data Research Laboratory)** on the EU-funded **STORAIGE** initiative, owning the full data science lifecycle from EDA and feature engineering to model evaluation for real-time decision-support use cases.
- Conducted **exploratory data analysis** and statistical profiling on **10,000+ records**, engineering time-aware and statistical features (rolling windows, z-scores) to improve signal quality for downstream modeling.
- Designed and benchmarked **anomaly detection models** combining supervised (Random Forest, Gradient Boosting) and unsupervised (Isolation Forest, One-Class SVM) approaches; applied **stratified cross-validation, hyperparameter tuning, and error analysis** on imbalanced data, with model selection driven by precision/recall/ROC-AUC.
- Delivered measurable business impact: **15% reduction in error rate** and **25% improvement in operational efficiency** through more accurate, faster anomaly signaling in real-time workflows.

PROJECTS

Ablacım , *Co-Founder & CTO* 12/2025 – Present
- Architected an AI-powered self-discovery platform combining **multimodal computer vision** (hand/face signal extraction) with **knowledge-grounded RAG** over curated rule bases (dream interpretation, numerology, ebced) to deliver personalized, multi-step conversational analyses with structured reports, quality checks, and privacy-first design.

TEKNOFEST and European Rover Challenge, *Autonomous Ground Rover* 09/2024

- **Finalist (Top 15 teams) at TEKNOFEST**; also competed in the **European Rover Challenge (ERC)**. Led the development of AI, ML algorithms, and computer vision systems for an autonomous ground rover. Utilized **ROS** and **Jetson Orin Nano** to enhance navigation, object detection, and decision-making, achieving **1,000+ successful detections** in evaluations.

ING Hubs Türkiye Datathon, *Customer Churn Prediction* 11/2025

- **Ranked Top 15%** among all participants in a competitive churn prediction challenge.
- Built end-to-end churn prediction pipeline with custom evaluation metric (composite **Gini + Recall + Lift**).
- Performed EDA on imbalanced customer data, engineered features from transactional behavior.
- Applied **XGBoost/LightGBM** with cross-validation and hyperparameter tuning.

DetectHub, *Computer Vision Project* 04/2024

- Founded **DetectHub**, a collaborative platform for 500+ researchers and developers in computer vision and AI, enabling the creation of 200+ models and 300+ datasets, improving object detection accuracy by 15% and accelerating project timelines.

SKILLS

Data Science & Analytics: — Python (Pandas, NumPy, SciPy), SQL (PostgreSQL, MS SQL) • **ML/DL:** — Scikit-Learn, XGBoost/LightGBM, PyTorch, TensorFlow, Keras, Cross-validation, Hyperparameter Tuning, Computer Vision • **Visualization:** — Matplotlib, Seaborn, Plotly, Jupyter • **GenAI/NLP:** — LangChain/LangGraph, RAG, LLM Fine-tuning (SFT), Embeddings, Vector DBs • **Engineering & MLOps:** — FastAPI, Docker, Kubernetes, Git/GitLab CI/CD • **C (Programming Language)**

LANGUAGES

Turkish: *Native* • **English:** *Fluent*

INTERESTS

Aikido: (Shodan: Officially Licensed Black Belt) • **Science** • **Space and Technology** • **Book and Authorship** • **Art and Design**